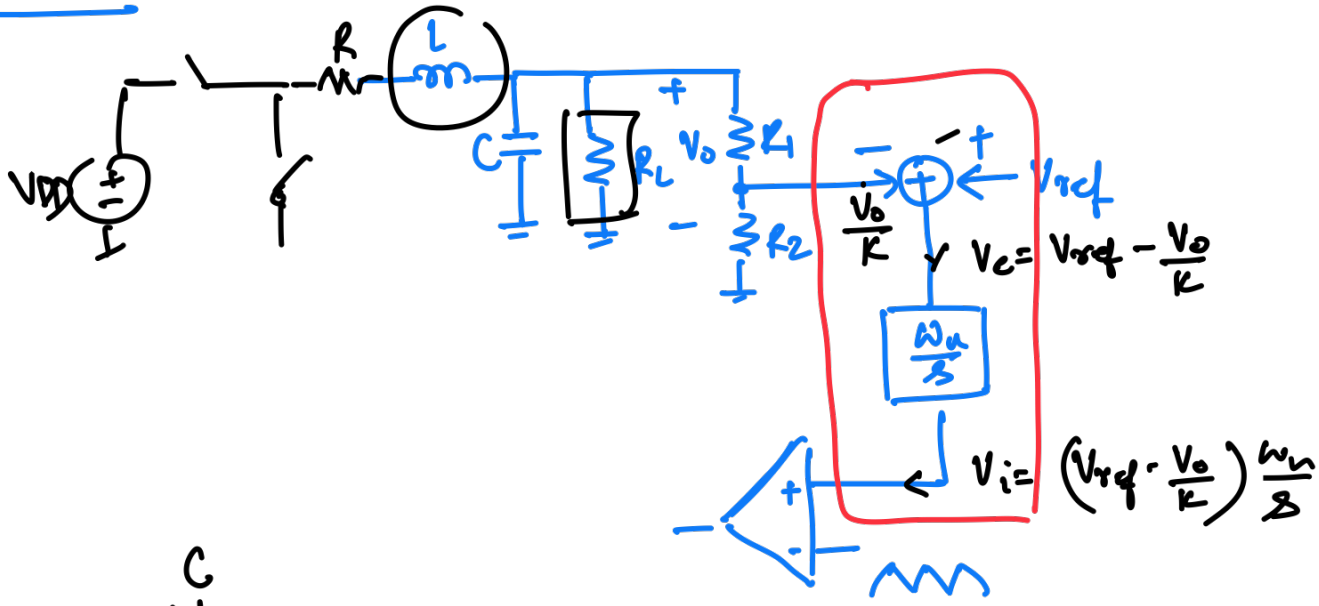
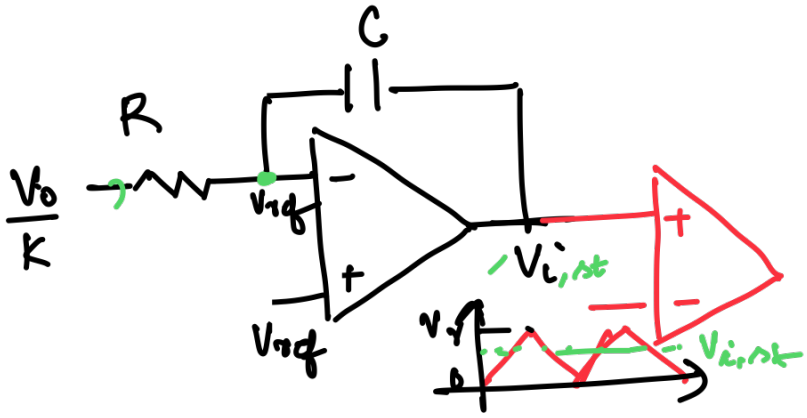
Lecture #26





$$\left(\frac{V_o}{K} - V_{ref}\right)\frac{1}{R} = (V_{ref} - V_i)\,sC$$

$$V_i = \left(\frac{V_o}{K} - V_{ref}\right)\frac{1}{sRC} + V_{ref}$$

$$= \left(\frac{V_o}{K} - V_{ref}\right)\frac{\omega_u}{s} + V_{ref}. \xrightarrow{t \to \infty} V_{i,st}$$

5V → Buck Converter → 3.3V, 2.5

3.3V → LDO → 2.5V

5V → Buck Converter → 1.8V → LDO → 1.2

→ LDO → 1.0

0.9

Analog filters.



- Audio signal (20 Hz – 20 KHz) — [ ] — V/I (20 Hz – 40 KHz) Analog.

40 KHz.

19th.

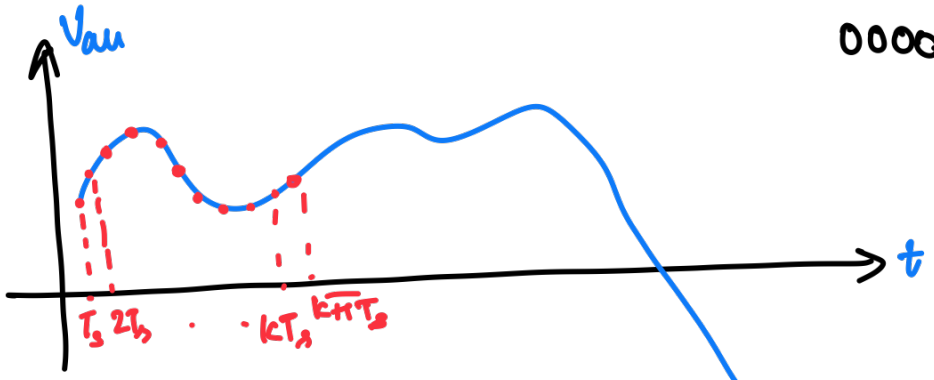
10101110

25th ← 25th Mar

9:00 ← @ 3:00 pm

0000 ← 10101110..

0000

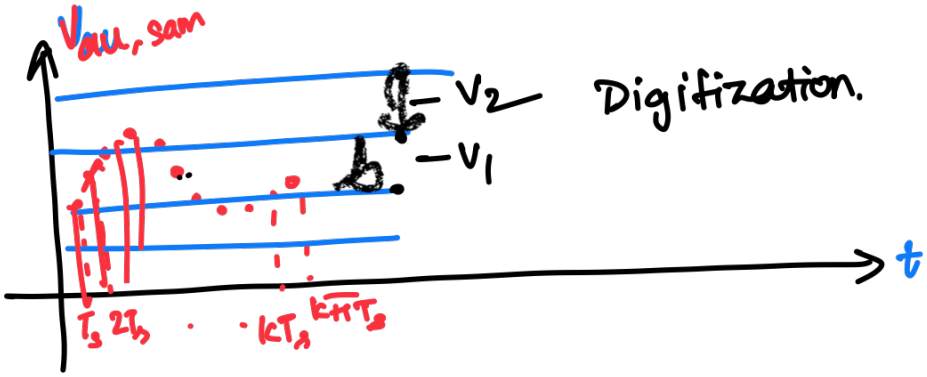


(ADC)

Analog-to-Digital converters.

Sampling + Digitization

10101110

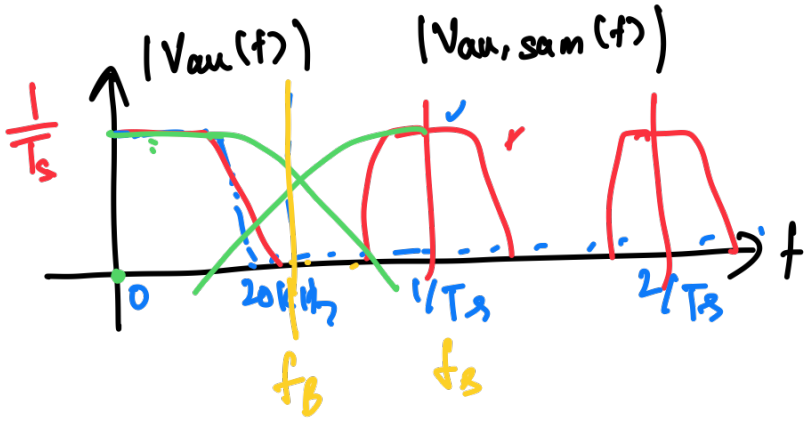


Digitization.

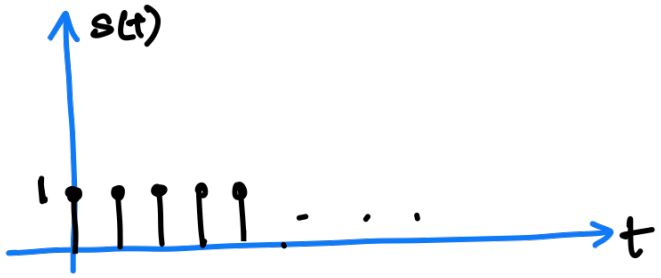
Analog.

(Digital-to-Analog Conv.)

(DAC)



Van x



$= V_{an,sam}$

$$s(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s) = \sum a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t)$$

$$\delta(t - kT_s) = \begin{cases} 1, & t = kT_s \\ 0, & \text{otherwise} \end{cases}$$

$$V_{an,sam}(f) = V_{an}(f) * S(f)$$

$$s(t) = \sum a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t), \quad \text{where } \omega_s = \frac{2\pi}{T_s}$$

$$\int_0^{T_s} s(t) \cdot \cos(n\omega_s t)\, dt = a_n \int_0^{T_s} \cos^2(n\omega_s t)\, dt$$

$$\int_0^{T_s} \delta(t) \cos(n\omega_s t)\, dt = a_n \int_0^{T_s} \frac{1 - \cos(2n\omega_s t)}{2} dt$$

$$1 = \frac{a_n}{2} \cdot T_s$$

$$a_n = \frac{2}{T_s}, \quad b_n = 0$$

$$s(t) = \sum \frac{2}{T_s} \cos(n\omega_s t)$$

$$= \sum \frac{2}{T_s} \frac{e^{jn\omega_s t} + e^{-jn\omega_s t}}{2}$$

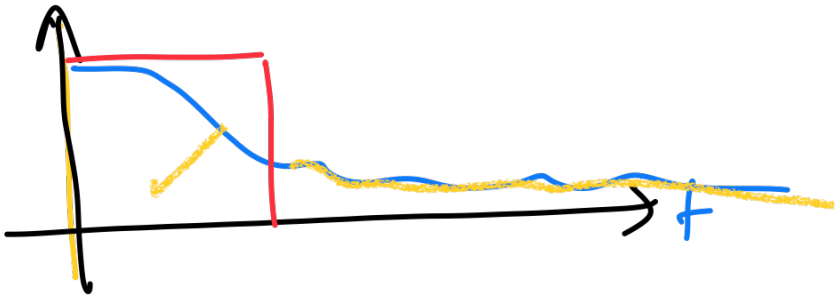
$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn\omega_s t} \quad \checkmark \xrightarrow{S(f)} \frac{1}{T_s} \sum \delta(\omega - n\omega_s)$$

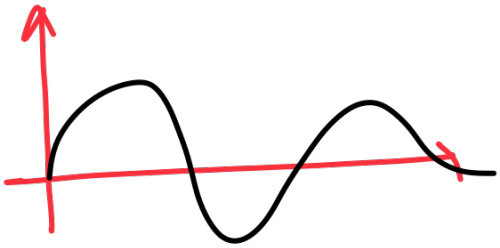
$$V_{avg\,sam}(\omega) = V_{av}(\omega) * \frac{1}{T_s}\sum \delta(\omega - n\omega_s)$$

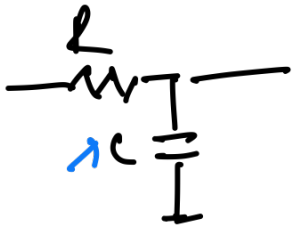
$$= \frac{1}{T_s}\sum V_{av}(\omega - n\omega_s)$$

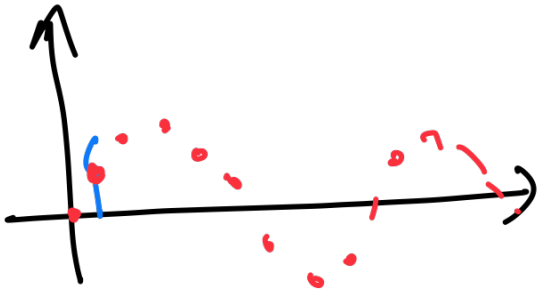
$$V_{av,\,sam}(f) = \frac{1}{T_s}\sum V_{av}\left(2\pi f - n\frac{2\pi}{T_s}\right)$$

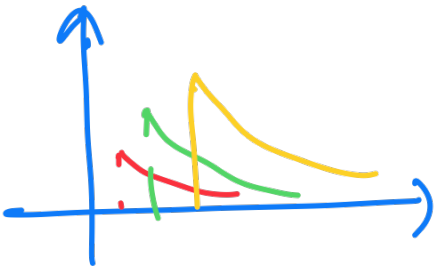
$$= \frac{1}{T_s}\sum V_{av}\left(f - \frac{n}{T_s}\right)$$

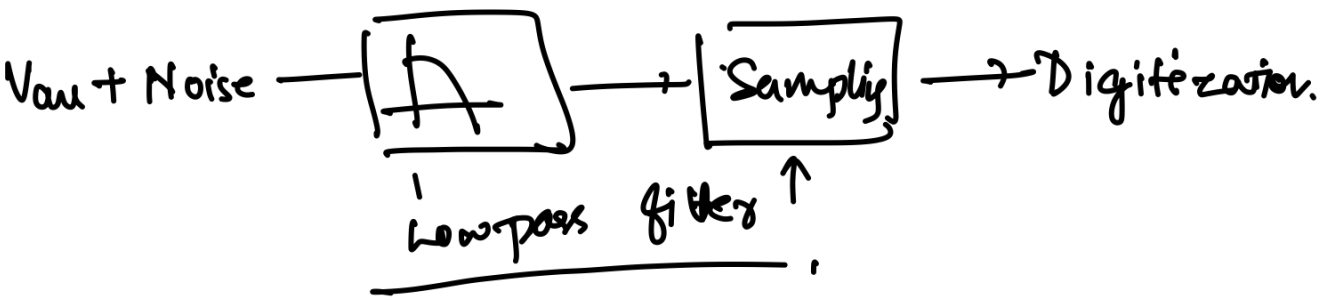
















Bandpass filter.